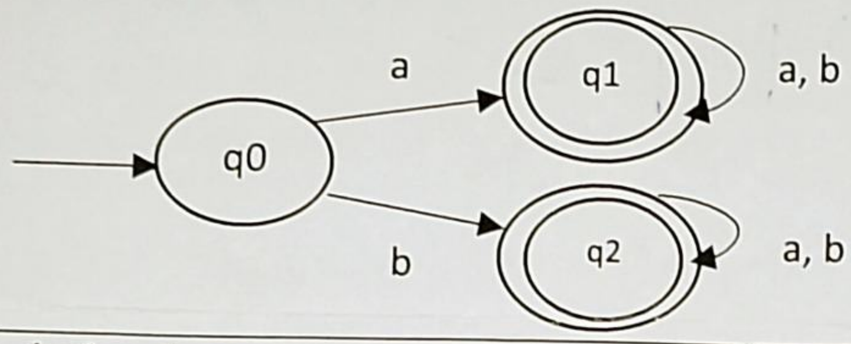
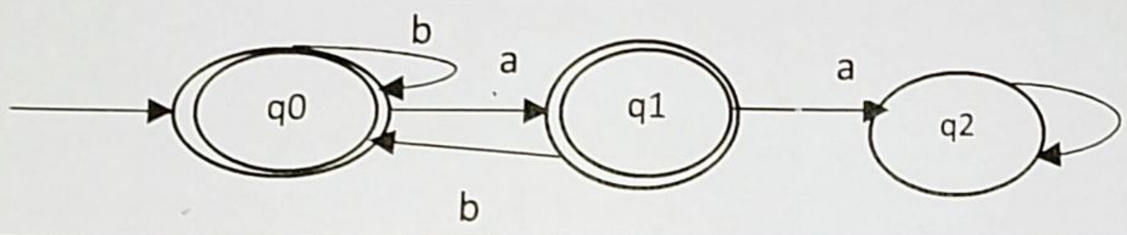
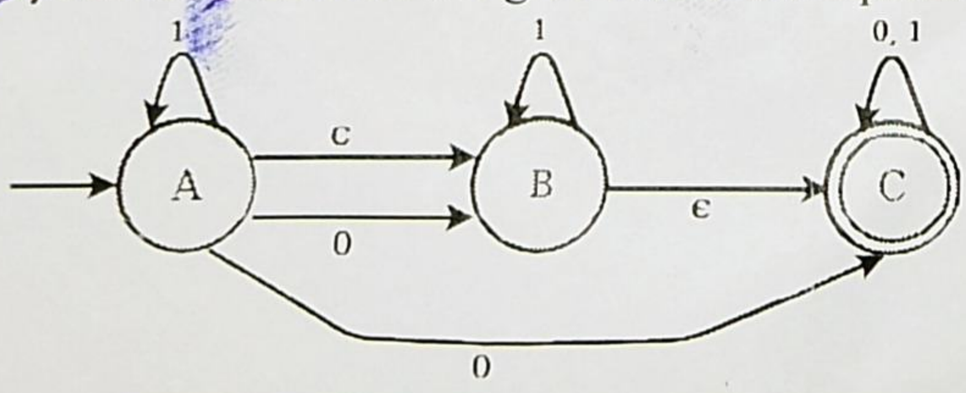
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING(A)
Osmania University, Hyderabad-07
CLASS TEST-1

SUBJECT: Automata Languages and Computation
Class: B.E(CSE)
Semester: IV

Date: XX-03-2026
Max. Marks: 30
Duration: 1.5 Hours (12.00 PM - 1:30 PM)

S.No.	PART-A Answer All Questions(5x2=10M)	Marks	BT	CO
1. a)	For given Language L = $\{a^n b^{3n-1}, n>=1\}$, prove L is a Non-Regular Language using Pumping Lemma.	2	3	2
b)	Generate the Regular expression for the given FA. 	2	2	3
c)	Find the Language accepted by the given FA? Show string acceptancy for corresponding Language. 	2	2	3
d)	Construct Moore Machine which generates 0 for each a and 1 for each b in the input string from input alphabet {a, b}.	2	3	1
e)	Define CFG. Construct Left Most Derivation and Right Most Derivation for string "abbabb" from below CFG. S→AA A→aB B→bB / ε	2	2	3
	PART-B **Answer any Two Questions(10X2=20M)**			
2.	a)Construct DFA on Σ = {a, b}, which accepts strings start with 'ab' must end with 'aa'. Show the string acceptancy for one accepted and nonaccepted string. b)Construct NFA for the given NFA-with epsilon. 	5 5	4 3	1 3
3.	a) Construct PDA for $L=\{a^n b^m c^m d^n, n,m >=1\}$. Perform string acceptancy with two sample strings. b) Construct the PDA for the following CFG S → 0A/AB A → 0AB/1 B → 1	6 4	4 3	1 3

(P.T.O.)

4.	a)Minimize the given FA and show the equivalence of two FA's.		5	3	1

States	0	1
→q0	q1	q2
q1	q2	q4
q2*	q0	q1
q3	q2	q4
q4*	q0	q3

	b)Construct CFG for the following PDA $\partial(q_0, a, z_0) \rightarrow (q_0, az_0)$ $\partial(q_0, a, a) \rightarrow (q_0, ba)$ $\partial(q_0, b, a) \rightarrow (q_1, \varepsilon)$ $\partial(q_1, b, b) \rightarrow (q_1, b)$	5	3	3

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- DFA ✓
- NFA
- Mealy/Moore ✓
- NFA with ∈ without ∈ ✓
- RE ✓
- DFA to RE ✓
- LR parser
- DFA to RE ✓
- RE to DFE ✓
- RE to NFA with ∈ ✓
- minization (2) ✓
- PDA
- PDA ✓
- CFL → P.L → Properties (2D) ✓
- L → P.L → properties (C+D) ✓
- Machine → U-IV ✓
- CNF to GNF ✓
- Red. of CFL ✓
- ammar, D Tree ✓
- biguity ✓

→ Chomsky hierarchy ✓

→ CSL ✓

→ PCP ✓